THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

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CLASS 9 — SCIENCE · MEDIUM LEVEL CHAPTER TEST SET 1

Chapter 8: **Sound Waves: Characteristics and Applications** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The frequency of a wave is 500 Hz and the speed is 340 m/s. The wavelength is: 1

(a) 170000 m **(b)** 1.47 m **(c)** 0.68 m **(d)** 0.0015 m

2. Sound of two different tuning forks has frequencies 256 Hz and 512 Hz. The wavelength is: 1

(a) zero **(b)** greater for the 256 Hz fork
(c) greater for the 512 Hz fork **(d)** equal

3. Two musical instruments produce notes of the same pitch and loudness but you can tell them apart. This is due to differing: 1

(a) speed **(b)** frequency
(c) amplitude **(d)** quality or timbre

4. A sound wave has a frequency of 2000 Hz and a speed of 340 m/s. The number of waves in a 3.4 m length is: 1

(a) 20 **(b)** 2 **(c)** 40 **(d)** 10

5. A wave of wavelength 2 m and speed 340 m/s has a time period of: 1

(a) 680 s **(b)** 170 s **(c)** 0.00588 s **(d)** 0.0059 Hz

6. If the amplitude of a sound wave is doubled, the loudness: 1

(a) stays same **(b)** decreases **(c)** increases **(d)** becomes zero

7. In a sound wave, the distance between a compression and the next rarefaction is: 1

(a) two wavelengths **(b)** one wavelength
(c) zero **(d)** half a wavelength

8. The loudness of sound is measured in: 1

(a) hertz **(b)** decibels **(c)** metres **(d)** watts

9. A tuning fork of 512 Hz produces sound of wavelength 0.67 m. The speed is about: 1

(a) 756 m/s **(b)** 0.0013 m/s **(c)** 343 m/s **(d)** 512.67 m/s

10. If the frequency is doubled, the pitch: 1

(a) stays same **(b)** becomes zero **(c)** becomes higher **(d)** becomes lower

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. The frequency of a wave is 20 Hz and its wavelength 17 m. Find the speed and the time for 10 oscillations. 2

12. Why is the voice of a person who has inhaled helium squeaky? 2

13. How do you get the speed of sound with a gun and a stopwatch method? Describe in two points. 2

14. A man standing 51 m from a cliff claps. After how much time will he hear the echo (v = 340 m/s)? 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. A source produces 4000 vibrations in 10 s. Find the frequency, the time period and the wavelength in air (v = 340 m/s). 3

16. Calculate the time interval between a lightning flash and the thunder if a lightning strikes 3.4 km away (v = 340 m/s). 3

17. Explain the working of a megaphone and a hearing aid. 3

OR

A ship sends an ultrasonic signal and receives the echo from the sea bed 2 s later. Find the depth (v = 1500 m/s) and the time if the depth were doubled. 3

18. A bat emits an ultrasonic pulse of 50 kHz. Find the wavelength in air (v = 340 m/s) and the time for the echo from an insect 1.7 m away. 3

OR

State three factors affecting the loudness of sound heard by a listener. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. A person standing between two parallel cliffs claps. He hears the first echo after 1 s and the second after 2 s. Find the distance between the cliffs and the distances from the person (v = 340 m/s). What would be the time if he is at the centre? 5

OR

Explain, with graphs described in words, how two sounds with different amplitudes and frequencies differ. Sketch the case for A (amplitude 2, frequency 100 Hz) and B (amplitude 1, frequency 200 Hz). 5

20. Explain the human ear and how the signal travels from the outer ear to the brain. State two precautions to protect hearing. 5

OR

Calculate the frequency and the wavelength of the sound waves heard by a driver if a horn gives 512 Hz in air at 20 degrees Celsius (v = 344 m/s), and the new wavelength at 40 degrees Celsius (v = 356 m/s). 5

END OF QUESTION PAPER — **Total: 40 Marks**